



Assignment # 2

Hide & Seek

1 Introduction

This assignment involves implementing a Hide and Seek game, exploring game theory concepts, and optimal strategies.

2 Game Description

The game is played on an N-sized linear world, where two players, a hider and a seeker, compete.

- **Hider:** Chooses a location in the world to hide.
- **Seeker:** Attempts to find the hider by choosing a location in the world to explore.

2.1 Scoring Mechanism

There are three types of places in this world

1. Hard places for the seeker: the seeker gets higher points upon winning, while the hider gets lower points upon winning
2. Neutral places: both hider and seeker get the same amount of points upon winning.
3. Easy places for the seeker: the seeker gets lower points upon winning, and the hider gets higher points upon winning

The following table represents an example of the hider's payoff matrix for a 4-place world. The rows represent the hider's move.

1. Place 3 is a hard place for the seeker
2. Place 1 is a neutral place where the score = 1
3. Places 2 and 4 are easy places for the seeker

	S_1	S_2	S_3	S_4
H_1	-1	1	1	1
H_2	2	-1	2	2
H_3	1	1	-3	1
H_4	2	2	2	-1



3 Assignment Requirements

You are required to implement the Hide and Seek game with the following criteria:

- Create a linear world shape that has an input N , which represents the world size.
- Allow a human player to choose to be either the hider or the seeker, playing against the computer.
- Choose randomly the type of each place.
- Assign a score to each type following the mentioned criteria
- Formulate the game as a linear programming problem from the hider's or the seeker's point of view (depending on what the human chooses).
- Solve the Computer's LP (hider/seeker) and produce the probability of each strategy (move) using a simplex solver (built-in or your implementation).
- Visualize the game matrix, the probabilities, and the type of places using a GUI.
- Create an interactive mode where the human can play multiple rounds with the computer.
- Create a simulation mode where you create a random player (hider/seeker) and play the game for 100 rounds.
- Show the number of rounds won and the total score by the player and the computer.
- Create a button to reset the game.

4 Bonus

4.1 Proximity Score

The hider is penalized by being rewarded fewer points if the seeker's choice was near the hider's place. The hider's score increases as he gets away from the seeker. For example:

- If the seeker's location is one cell away from the hider's location, the hider's score is multiplied by 0.5
- If the seeker's location is two cells away from the hider's location, the hider's score is multiplied by 0.75
- otherwise the hider's score is multiplied by 1



4.2 2D World

Instead of applying the proximity score to a linear world, arrange the world as a 2D grid, i.e: 16 places on a line are now 4×4 .

5 Deliverables

You are required to submit:

1. Your code
2. A report containing an explanation of the concepts used, an explanation of your code, and some test cases.

You should work in groups of 3.